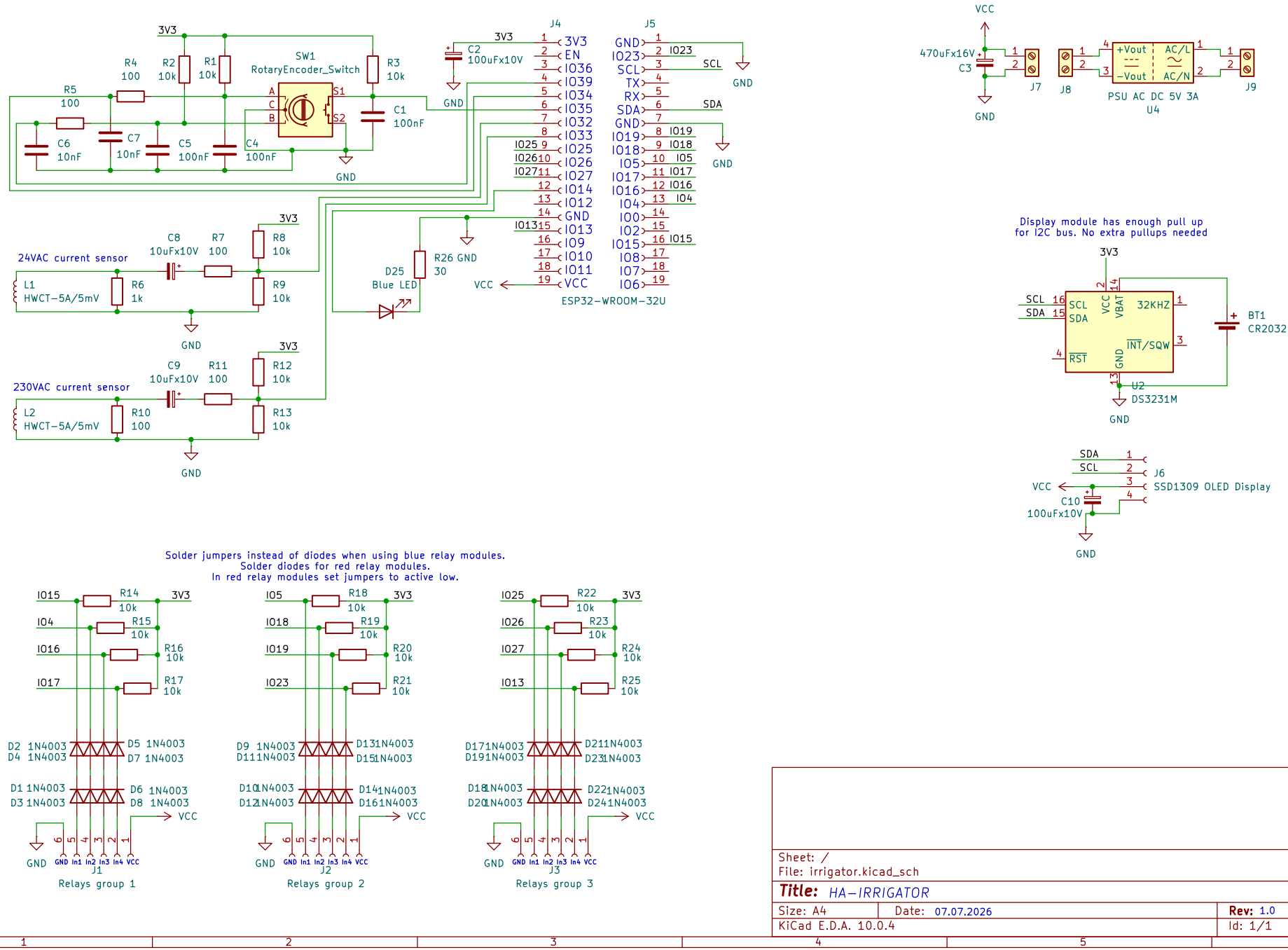

3V3
R4 100
R2 10k
R1 10k
SW1 RotaryEncoder_Switch
R3 10k
R5 100
A
C
B
S1
S2
C1 100nF
C6 10nF
C7 10nF
C5 100nF
C4 100nF
GND
J4
J5
3V3
C2 100uFx10V
GND
3V3
EN
IO36
IO39
IO34
IO35
IO32
IO33
IO25
IO26
IO27
IO14
IO12
GND
IO13
IO9
IO10
IO11
VCC
GND
IO23
SCL
TX
RX
SDA
GND
IO19
IO18
IO5
IO17
IO16
IO4
IO0
IO2
IO15
IO6
ESP32-WROOM-32U
24VAC current sensor
C8 10uFx10V
R7 100
R8 10k
L1 HWCT-5A/5mV
R6 1k
R9 10k
230VAC current sensor
C9 10uFx10V
R11 100
R12 10k
L2 HWCT-5A/5mV
R10 100
R13 10k
D25 Blue LED
R26 30
VCC
470uFx16V C3
J7
J8
+Vout
-Vout
AC/L
AC/N
PSU AC DC 5V 3A
U4
J9
Display module has enough pull up
for I2C bus. No extra pullups needed
SCL
SDA
VCC
VBAT
32KHZ
INT/SQW
RST
GND
BT1 CR2032
U2 DS3231M
SDA
SCL
VCC
C10 100uFx10V
J6 SSD1309 OLED Display
Solder jumpers instead of diodes when using blue relay modules.
Solder diodes for red relay modules.
In red relay modules set jumpers to active low.
IO15
IO4
IO16
IO17
R14 10k
R15 10k
R16 10k
R17 10k
D2 1N4003
D4 1N4003
D5 1N4003
D7 1N4003
D1 1N4003
D3 1N4003
D6 1N4003
D8 1N4003
GND In1 In2 In3 In4 VCC
J1
Relays group 1
IO5
IO18
IO19
IO23
R18 10k
R19 10k
R20 10k
R21 10k
D9 1N4003
D11 1N4003
D13 1N4003
D15 1N4003
D10 1N4003
D12 1N4003
D14 1N4003
D16 1N4003
J2
Relays group 2
IO25
IO26
IO27
IO13
R22 10k
R23 10k
R24 10k
R25 10k
D17 1N4003
D19 1N4003
D21 1N4003
D23 1N4003
D18 1N4003
D20 1N4003
D22 1N4003
D24 1N4003
J3
Relays group 3
Sheet: /
File: irrigator.kicad_sch
Title: HA-IRRIGATOR
Size: A4
Date: 07.07.2026
Rev: 1.0
KiCad E.D.A. 10.0.4
Id: 1/1